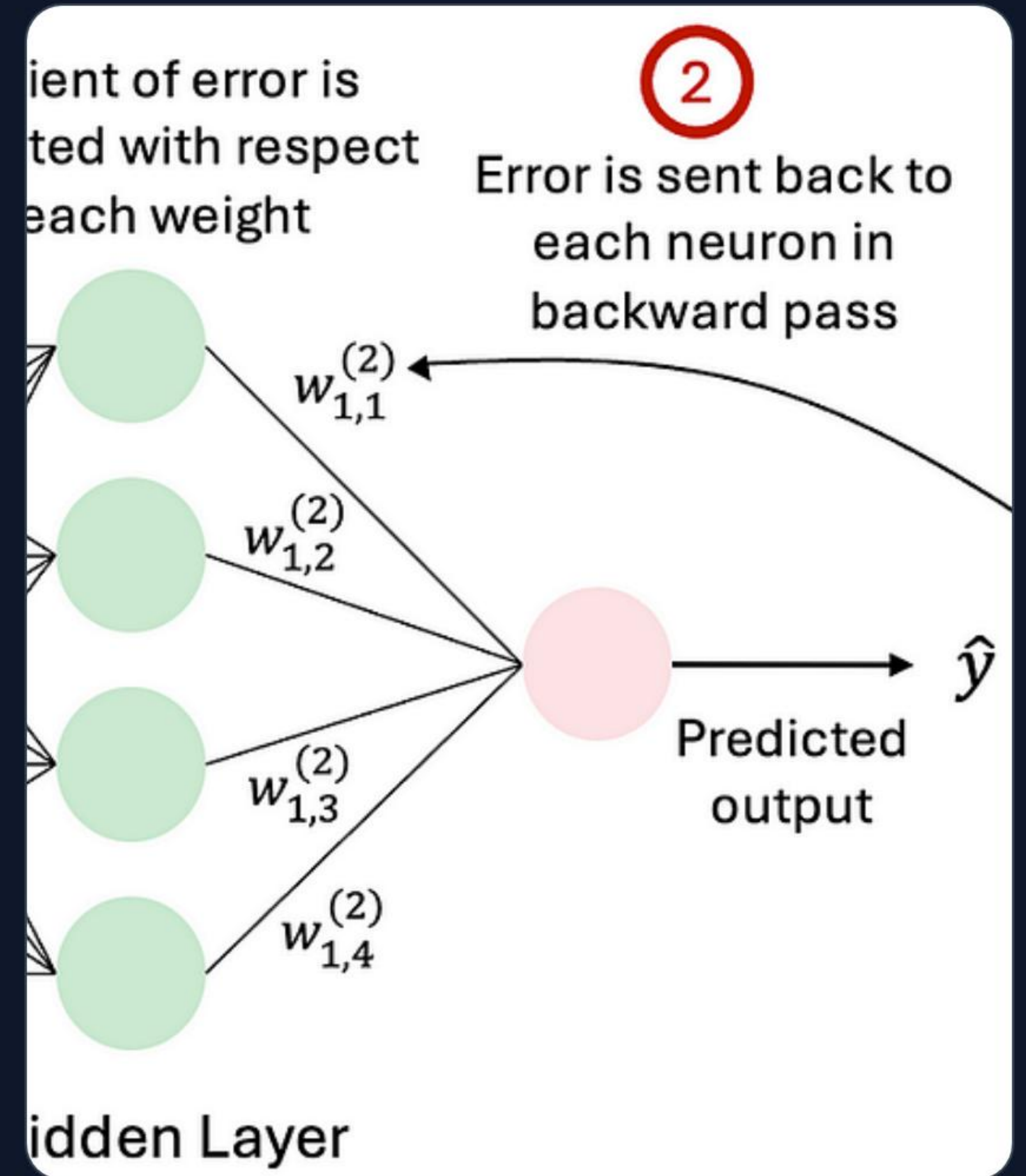


DEEP LEARNING OPTIMIZATION DYNAMICS

How AI Learns: Backpropagation i.e. Corrections

From initial mistakes to mathematical precision: how loss functions and gradient updates drive model optimization.



The 4-Step Training Cycle

STEP 01

Forward Pass

Input flows through layers to produce a prediction.



STEP 02

Loss Calculation

Loss function quantifies error against target.



STEP 03

Backpropagation

Chain rule calculates gradients backward through layers.



STEP 04

Weight Update

Optimizer adjusts parameters to reduce loss.

The Forward Pass: Generating a Guess

Inputs multiply by weights, pass through activations, and yield an initial prediction.

Linear Transformation

$$z = (w_1 \cdot x_1) + (w_2 \cdot x_2) + b$$

Inputs (x) scaled by adjustable weights (w) plus bias (b).

Non-Linear Activation

$$a = \sigma(z) \text{ OR } a = \text{ReLU}(z)$$

Enables network to model complex non-linear relationships.

Measuring the Mistake: The Loss Function

Quantifying discrepancy between prediction and ground-truth target.

REGRESSIONLOSS

Mean Squared Error (MSE)

$$L = \frac{1}{2} (y_{\text{target}} - \hat{y}_{\text{pred}})^2$$

Penalizes large deviations quadratically.

LLM / CLASSIFICATIONLOSS

Cross-Entropy Loss

$$L = -\log(p_{\text{target}})$$

Measures probability assigned to correct token.

Backpropagation: Applying the Chain Rule

Calculating partial derivatives backwards to attribute error to each weight.

The Chain Rule Equation

$$\frac{\partial L}{\partial w} = \left(\frac{\partial L}{\partial \hat{y}} \right) \cdot \left(\frac{\partial \hat{y}}{\partial z} \right) \cdot \left(\frac{\partial z}{\partial w} \right)$$

Decomposes total error change into step-by-step layer dependencies.

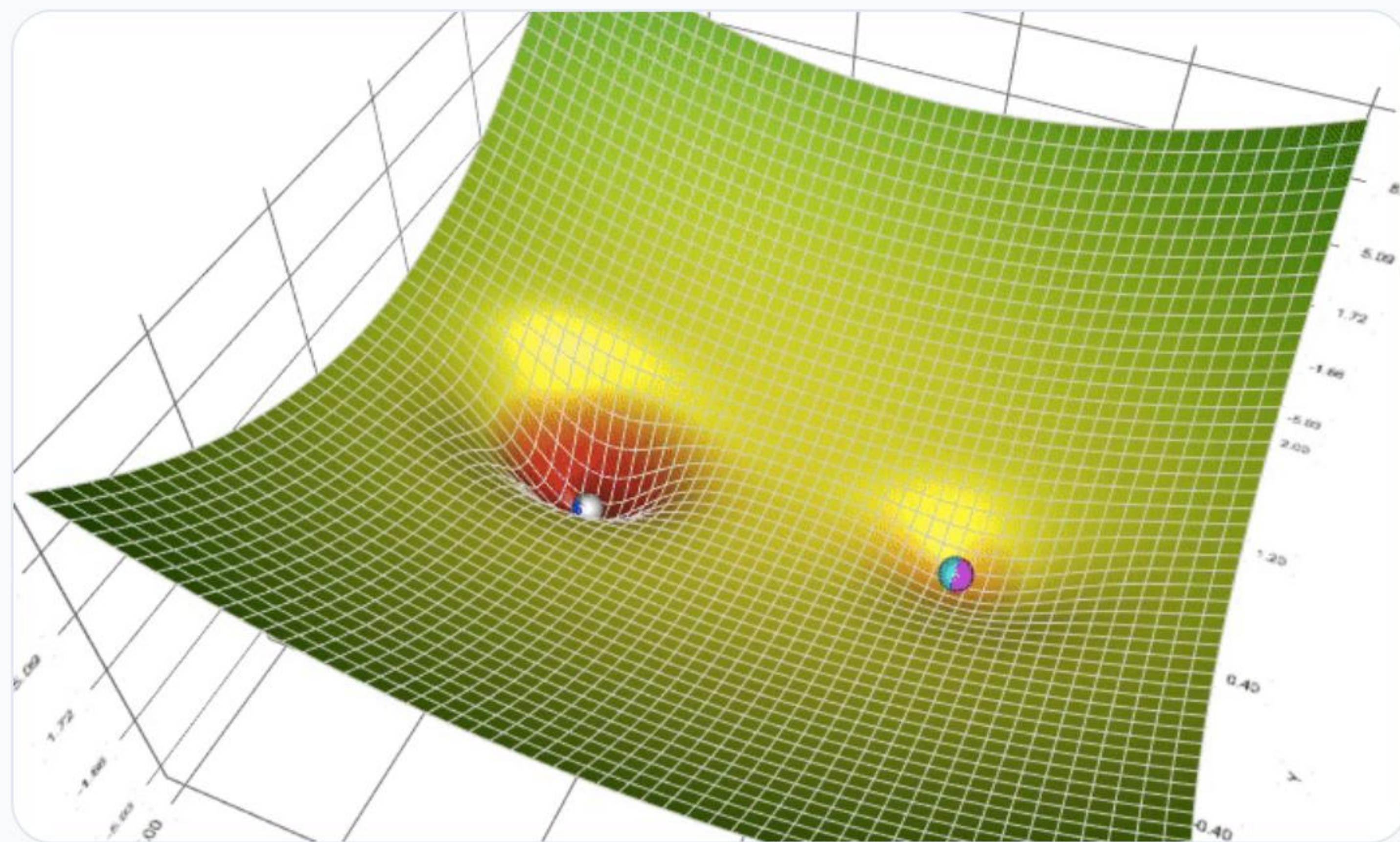
Key Insights

Gradient Direction: Sign indicates whether weight must increase or decrease.

Gradient Magnitude: Size determines how strongly weight influenced error.

Gradient Descent: Adjusting the Knobs

Stepping down the loss surface toward minimal prediction error.



Weight Update Formula

$$w_{\text{new}} = w_{\text{old}} - (\eta \times \partial L / \partial w)$$

η (Learning Rate): Hyperparameter scaling step size in direction of steepest descent.

CONCRETE LLM EXAMPLE

Token Correction: “The salmon swam near the ___”

Correcting logits to boost probability for target token “river”.

BEFORE CORRECTION

"vault": 62%

"river": 12% (Target)

"table": 8%

High Loss ($L = 2.12$)



BACKPROP & UPDATE

Gradient pushes weights connecting "salmon" and "swam" to boost "river" logit.



AFTER CORRECTION

"river": 84% (Target)

"vault": 4%

"table": 1%

Low Loss ($L = 0.17$)

The Goldilocks Problem: Learning Rate (η)

Step size dictates whether the model converges cleanly or fails to train.

TOO HIGH

Overshooting

Steps jump over optimal loss minimum, causing gradient divergence.

JUST RIGHT

Optimal Convergence

Steady reduction in loss toward global minimum with stable gradient flow.

TOO LOW

Stagnation

Takes millions of steps to make progress or gets trapped in local minima.

SCALE & TRAINING

From Random Noise to Coherent Intelligence

Repeating backpropagation across trillions of tokens transforms raw weights.

INITIAL STATE

Random Weights

Generates gibberish and high initial cross-entropy loss.



MIDTRAINING

Syntax & Structure

Learns basic grammar patterns and word co-occurrences.



CONVERGENCE

Contextual Reasoning

Captures subtle semantic, contextual, and relational concepts.

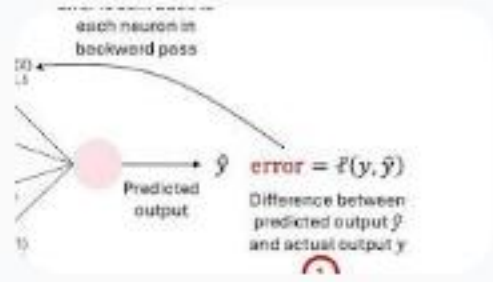


Key Takeaway

AI does not learn by magic—it learns by measuring error, assigning blame via the chain rule, and making tiny mathematical adjustments to billions of weights over time.

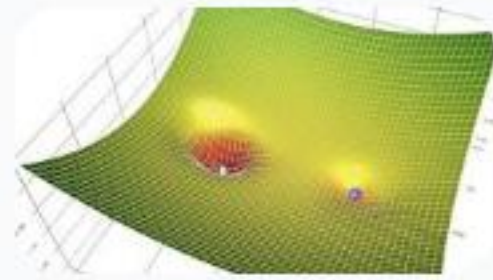


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